

6. (a) Nuclear fusion and nuclear fission reactions both produce heat energy. Describe and compare controlled nuclear fission and nuclear fusion reactions. [6 QER]

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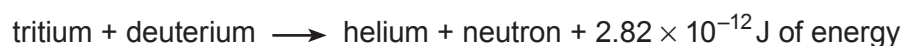
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- (b) A typical fusion reaction between tritium and deuterium is:



Calculate the number of fusion reactions needed each second to provide an energy output of 6.2 MJ every second. [2]

Number of reactions = .....



6. Nuclear decay in a fission reactor leads to the production of vast amounts of electrical power. The reactor has design features so there is sustained and controlled fission of uranium.

(a) Explain how the reactor is designed so that fission is sustained and controlled. [4]

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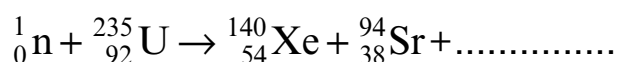
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- (b) A common pair of fission fragments from uranium-235 fission is xenon (Xe) and strontium (Sr). The nuclear equation for the reaction is shown below.



(i) **Complete** the equation. [2]

(ii) Explain why a nucleus of uranium-235 undergoes fission when it absorbs a neutron. [2]

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- (c) One product of the fission of uranium is strontium-94. Strontium-94 undergoes beta decay with a half-life of 75s into an isotope of yttrium (Y). Yttrium also undergoes beta decay and its half-life is 19 minutes. Its product is a stable isotope of zirconium.

(i) Write a balanced nuclear equation for the decay of strontium-94 into yttrium. [2]

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(ii) Explain what is meant by the statement *strontium-94 has a half-life of 75s*. [2]

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- (iii) Explain whether long-term safety precautions are required for the disposal of the waste product strontium-94. [2]

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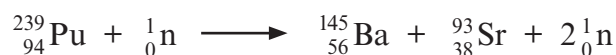


7. In a fast breeder nuclear reactor, uranium-238 ( ${}^{238}_{92}\text{U}$ ) is converted into the fissionable isotope plutonium-239 ( ${}^{239}_{94}\text{Pu}$ ). The reactor has control rods but no moderator.

The sequence, of the 3 stages, is shown.



- (a) State the total number of beta particles emitted during the 3 stages. .... [1]
- (b) When a plutonium-239 nucleus is bombarded with a **high-speed neutron** it splits into barium (Ba), strontium (Sr) and two high-speed neutrons.  
The reaction is shown below.



The two high-speed neutrons split more plutonium-239 nuclei and cause a nuclear fission chain reaction.

The fast breeder nuclear reactor does not require a moderator, but a conventional nuclear reactor does.

- (i) Explain the purpose of the moderator that is used in a conventional nuclear reactor. [2]

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- (ii) Suggest why the fast breeder nuclear reactor doesn't require a moderator. [1]

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- (c) High level radioactive waste (HLW) from nuclear power stations and nuclear medicine may be contained in solid glass (vitrified) and then kept securely in stainless steel lined, concrete containers in deep underground facilities. HLW is highly ionising and has a long half-life.

Discuss the advantages **and** disadvantages of storing HLW in this way. [6 QER]

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3. Nuclear fusion can take place in a fusion reactor.

- (a) Deuterium is one isotope of hydrogen, H, which consists of 1 proton and 1 neutron.  
Tritium is another isotope of hydrogen which consists of 1 proton and 2 neutrons.  
In one nuclear fusion reaction, deuterium and tritium undergo nuclear fusion to form helium, He, and one neutron.

Produce a **balanced nuclear equation** for the fusion of deuterium and tritium. [3]

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- (b) (i) Explain why nuclear fusion is difficult to achieve on Earth. [2]

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- (ii) One advantage of nuclear fusion compared to nuclear fission is that it doesn't produce radioactive waste.

State **two** reasons why radioactive waste is difficult to store safely. [2]

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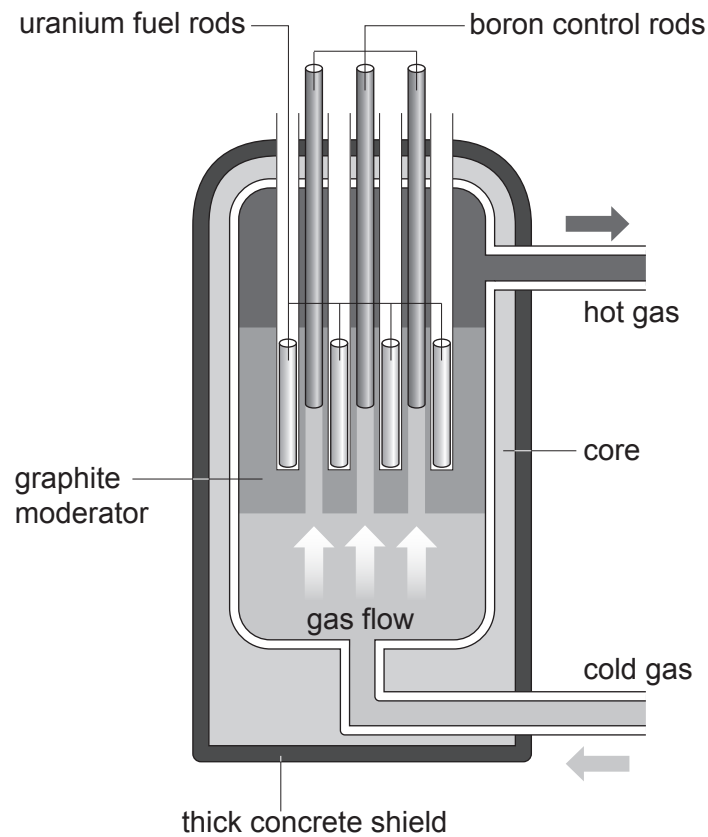
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6. Nuclear fission is a nuclear reaction that releases energy.  
The diagram shows a nuclear fission reactor.



- (a) In each fission reaction, on average three neutrons are released.

Explain how this can lead to an uncontrolled chain reaction **and** how the features of the reactor allow a controlled chain reaction to be achieved. [6 QER]

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(b) Another nuclear reaction is nuclear fusion.

One fusion reaction, which could provide energy on Earth in the future, involves two isotopes of hydrogen, H, deuterium and tritium.

Deuterium and tritium fuse to produce an isotope of helium, He, and one neutron.

(i) Use the information from the table below to complete the nuclear equation for this reaction. [3]

| Isotope   | Number of protons | Number of neutrons |
|-----------|-------------------|--------------------|
| deuterium | 1                 | 1                  |
| tritium   | 1                 | 2                  |



(ii) The main product of this reaction is helium.

Suggest why, if it can be achieved on Earth, nuclear fusion may be better for energy production than nuclear fission. [1]

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